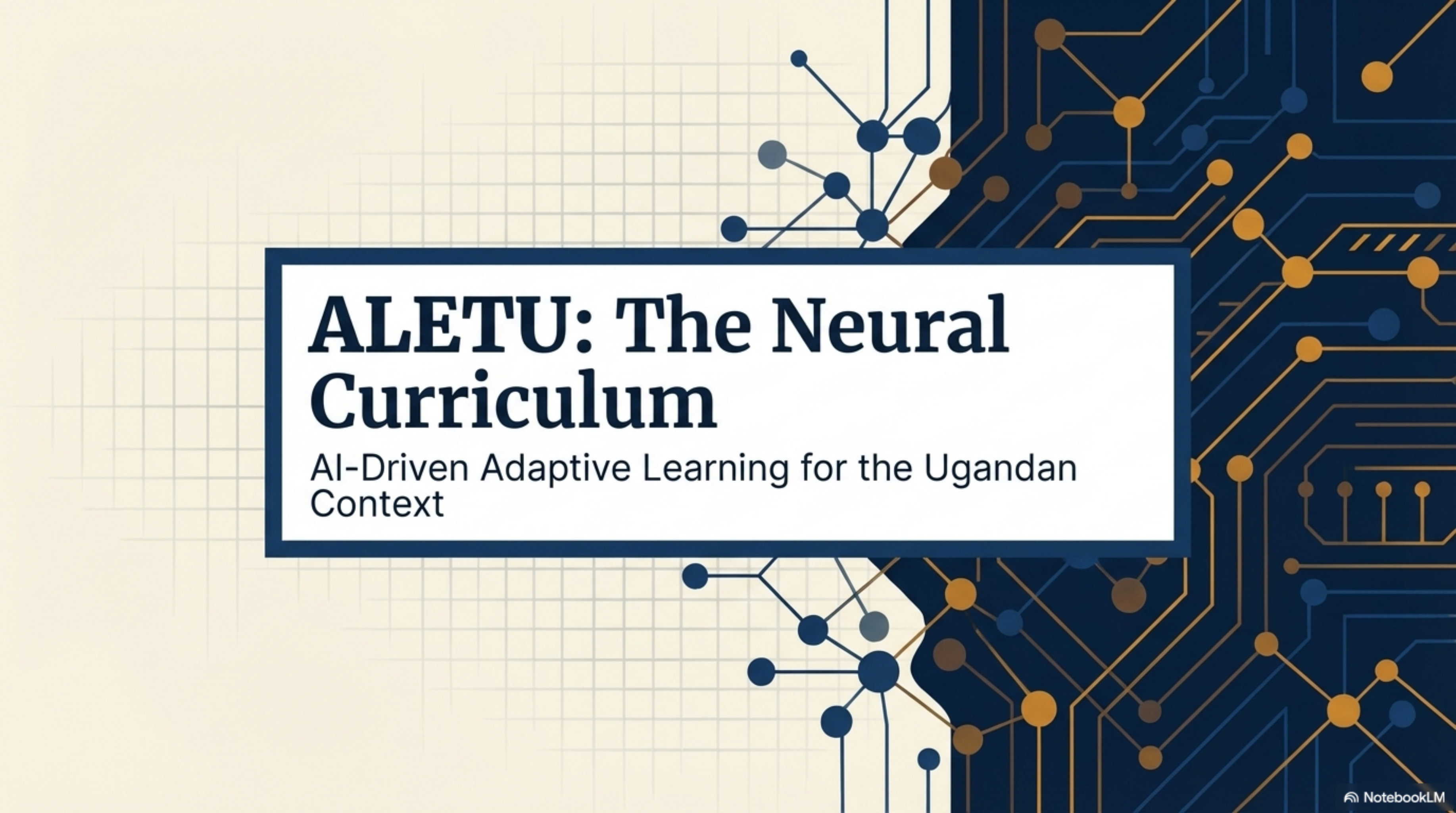


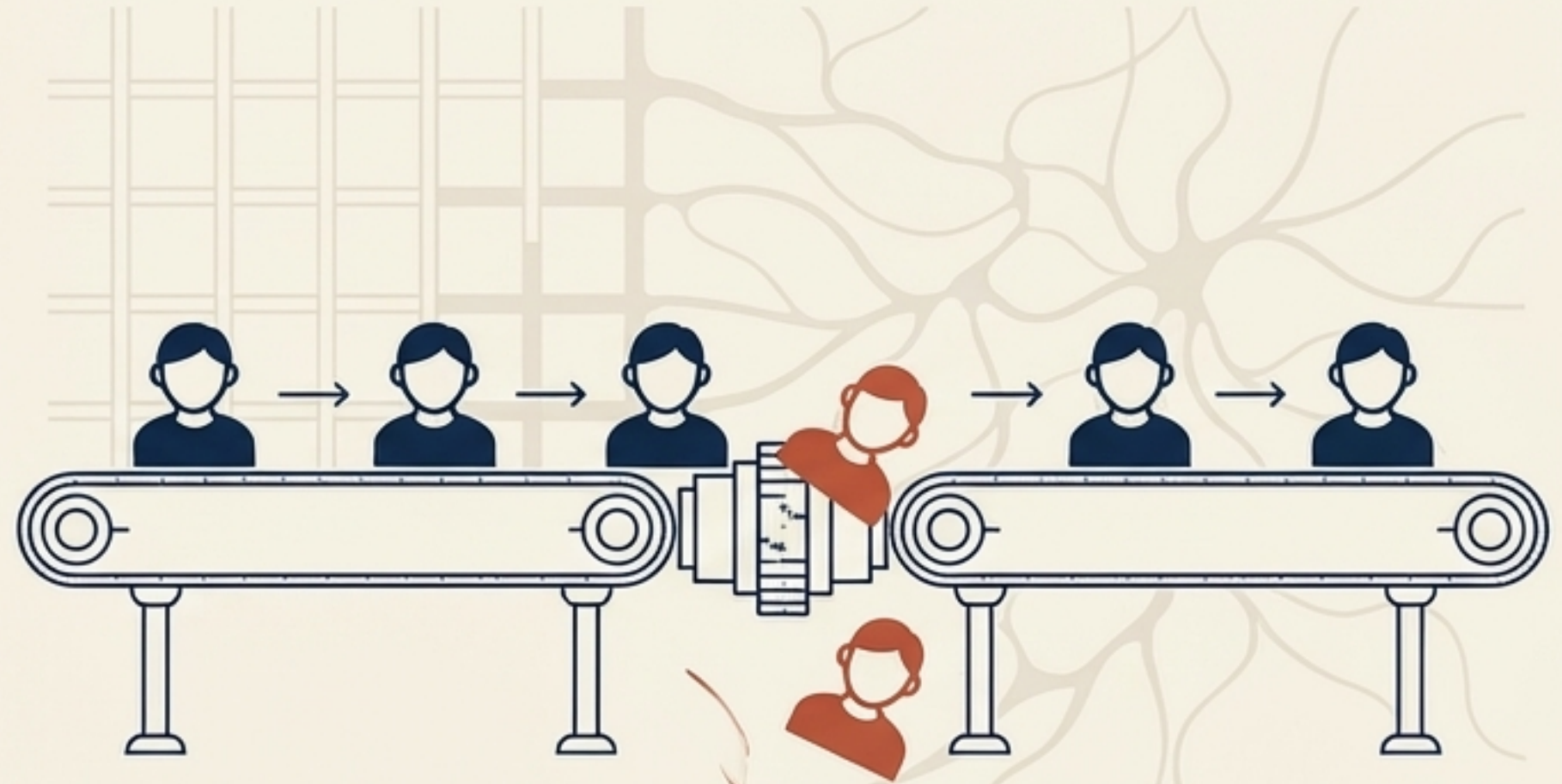


ALETU: The Neural Curriculum

AI-Driven Adaptive Learning for the Ugandan Context



The Friction of One-Size-Fits-All



17 Million

Additional
teachers required
in Africa by 2030

30 Million

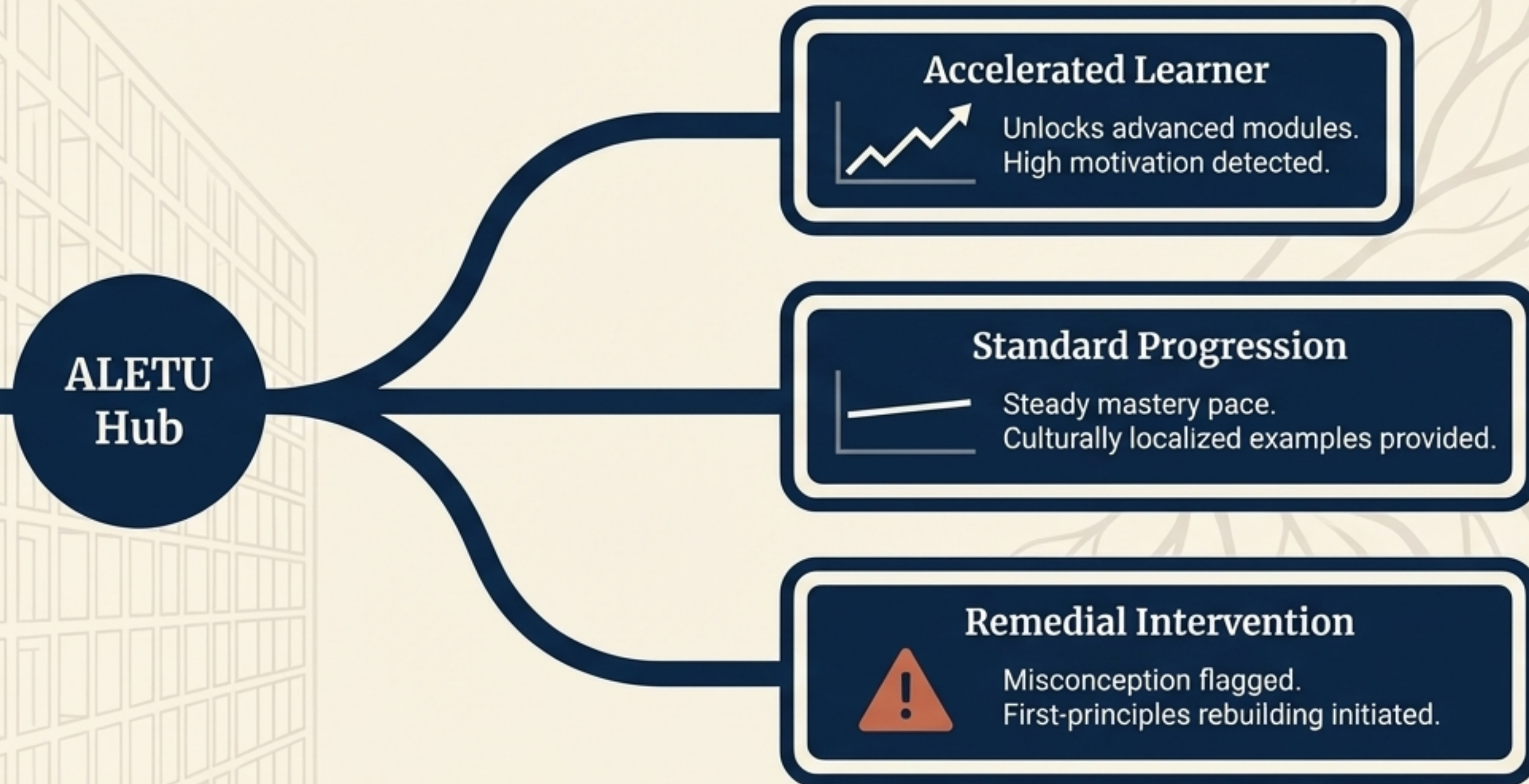
Primary-age
children out of
school in
Sub-Saharan Africa

Single Pace

Forces rote
memorization and
causes critical
knowledge gaps

A Paradigm Shift: Meet ALETU

ALETU transforms static syllabi into dynamic, highly personalized learning pathways. It simulates a responsive, culturally attuned tutor tailored specifically for the Ugandan National Curriculum Development Centre (NCDC) syllabus.

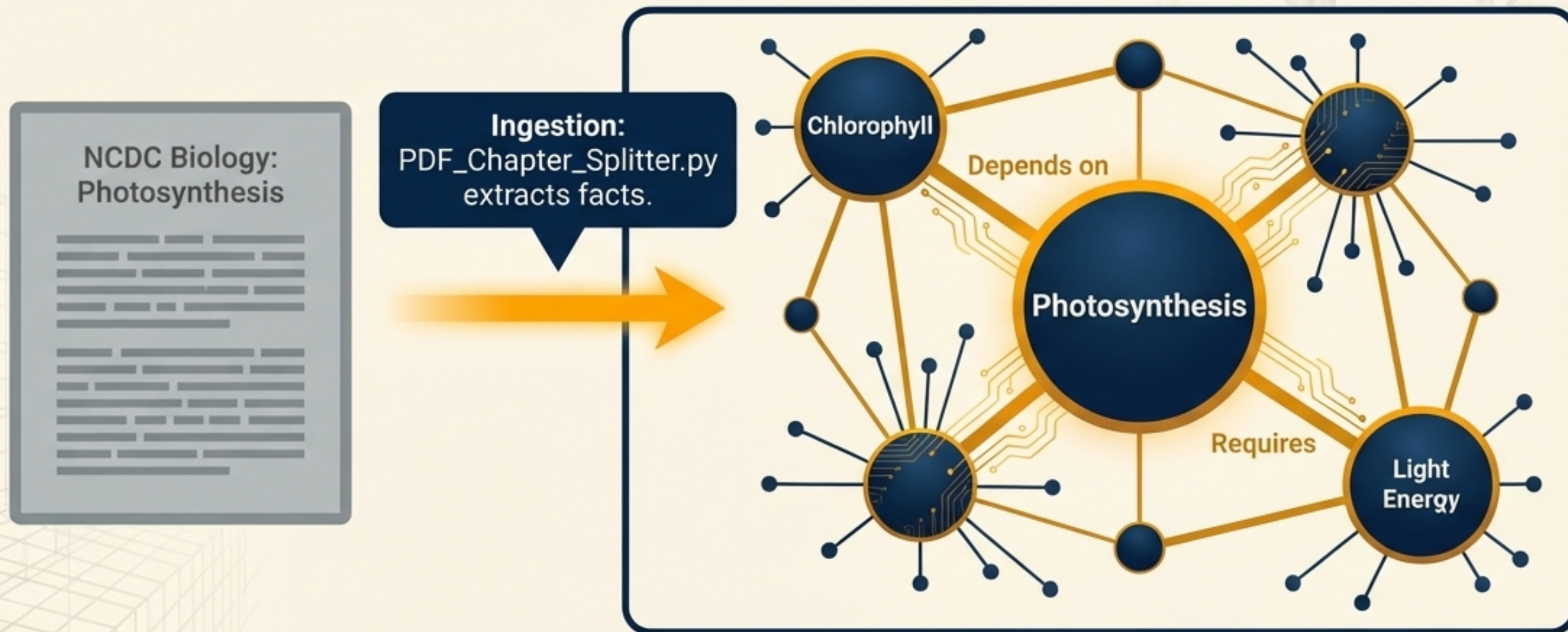


The EdTech Paradigm Shift Matrix

	Traditional Digital LMS	ALETU's Adaptive AI Ecosystem
Content Generation	Static, pre-recorded, finite.	Generative, infinite, tailored dynamically via LangChain .
Progression	Time-based (Week 1, Week 2).	Mastery-based via Bayesian Knowledge Tracing ($P \geq 0.9$).
Context & Culture	Western-default examples.	Grounded in Ugandan Knowledge Graphs (Parquet).
Analytics	Binary Pass/Fail grading.	Emotion tracking , motivation streaks, misconception analysis.

Core Mechanism 1: Knowledge Graphs

From Static PDFs to Living Data



Storage:

Hierarchical .parquet files (e.g., Biology.parquet).

Retrieval:

AI queries facts, ensuring absolute curriculum fidelity and zero hallucination.

Core Mechanism 2: Generative Content

Infinite, Culturally Relevant Lessons



System Mandate

The content_generator.py script operates under strict constraints to maintain a supportive, culturally resonant classroom tone, avoiding generic AI-speak.

Core Mechanism 3: Bayesian Knowledge Tracing (BKT)

The Science of Mastery

Variables Tracked

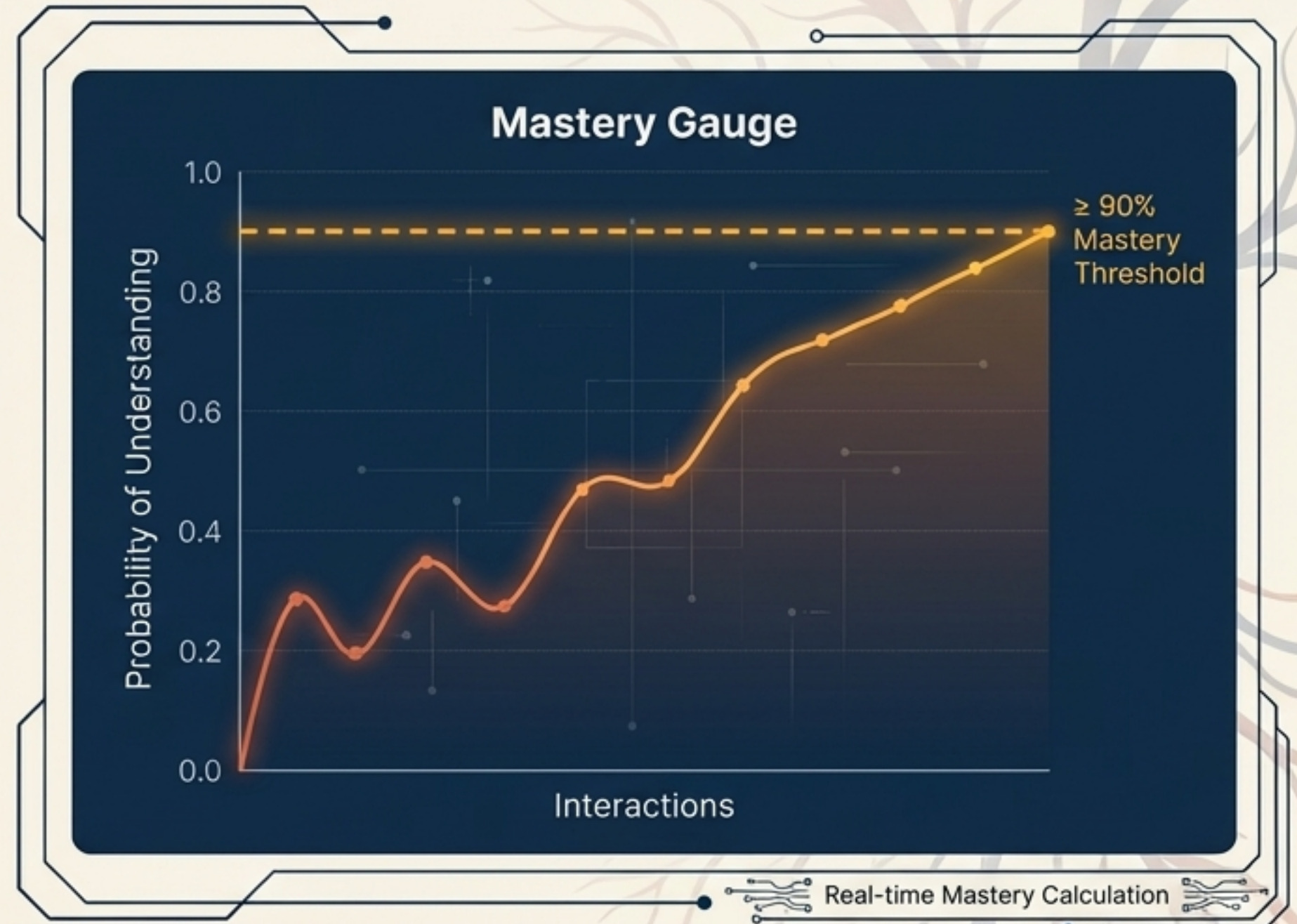
$P(\text{learn})$, $P(\text{slip})$, and $P(\text{guess})$ dynamically calculated per interaction.

The Goal

The curve must breach the 0.9 threshold before the AI unlocks the next curriculum subtopic.

Misconception Detection

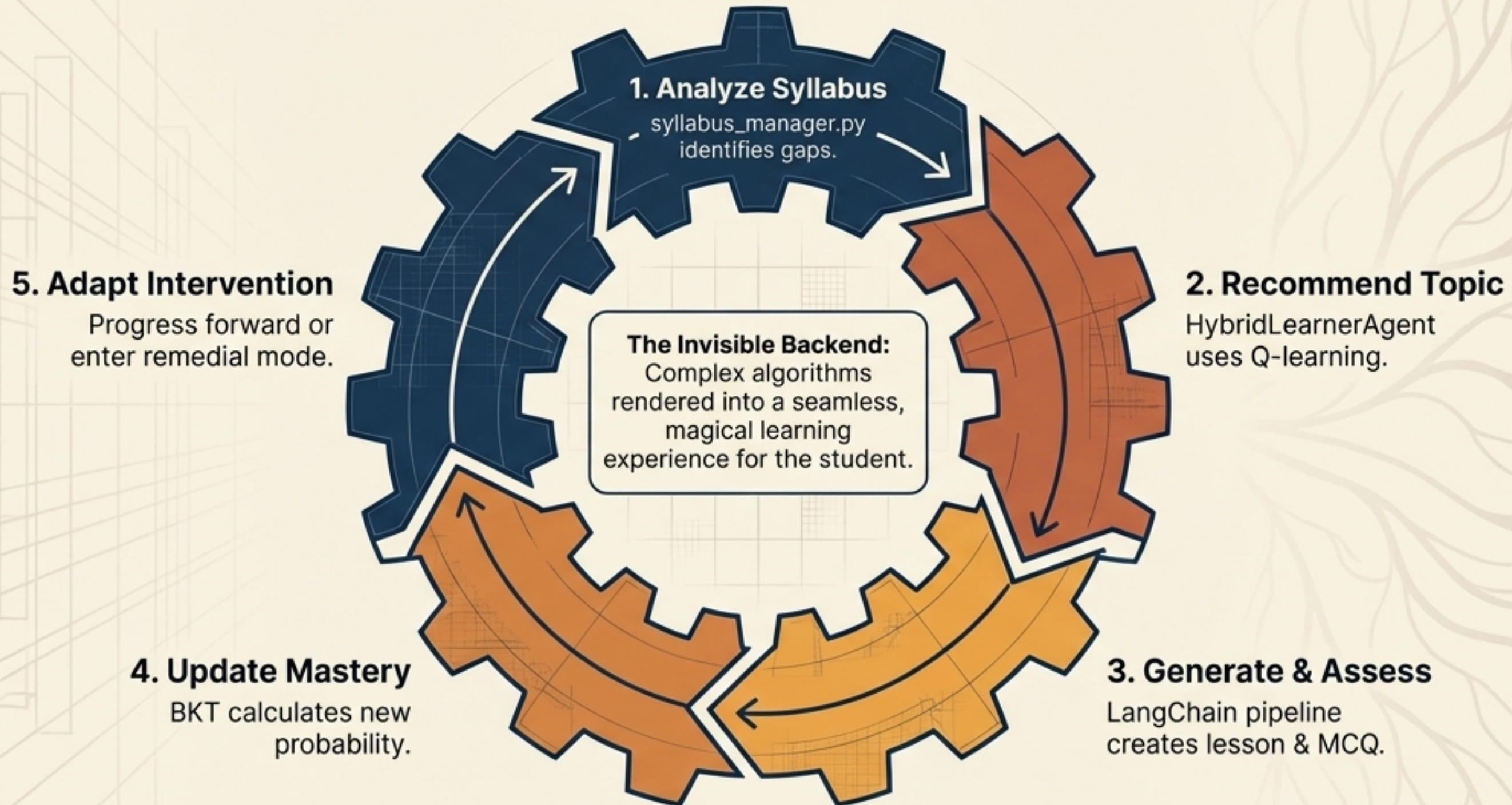
`mcq_analyzer.py` instantly flags errors and triggers tailored remedial interventions.



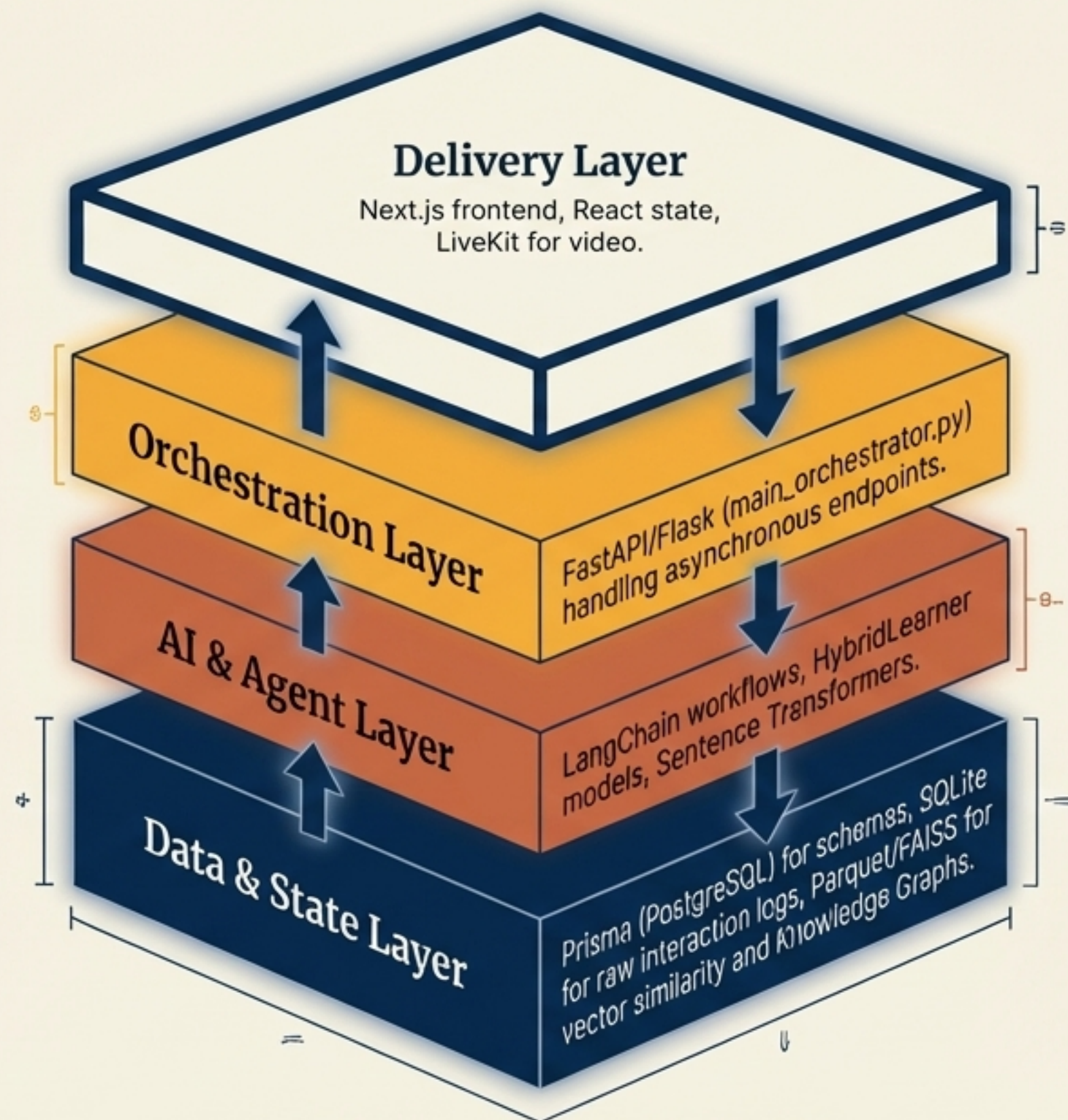
The Closed-Loop Architecture

From Syllabus to Mastery

The Closed-Loop Infinity Gear

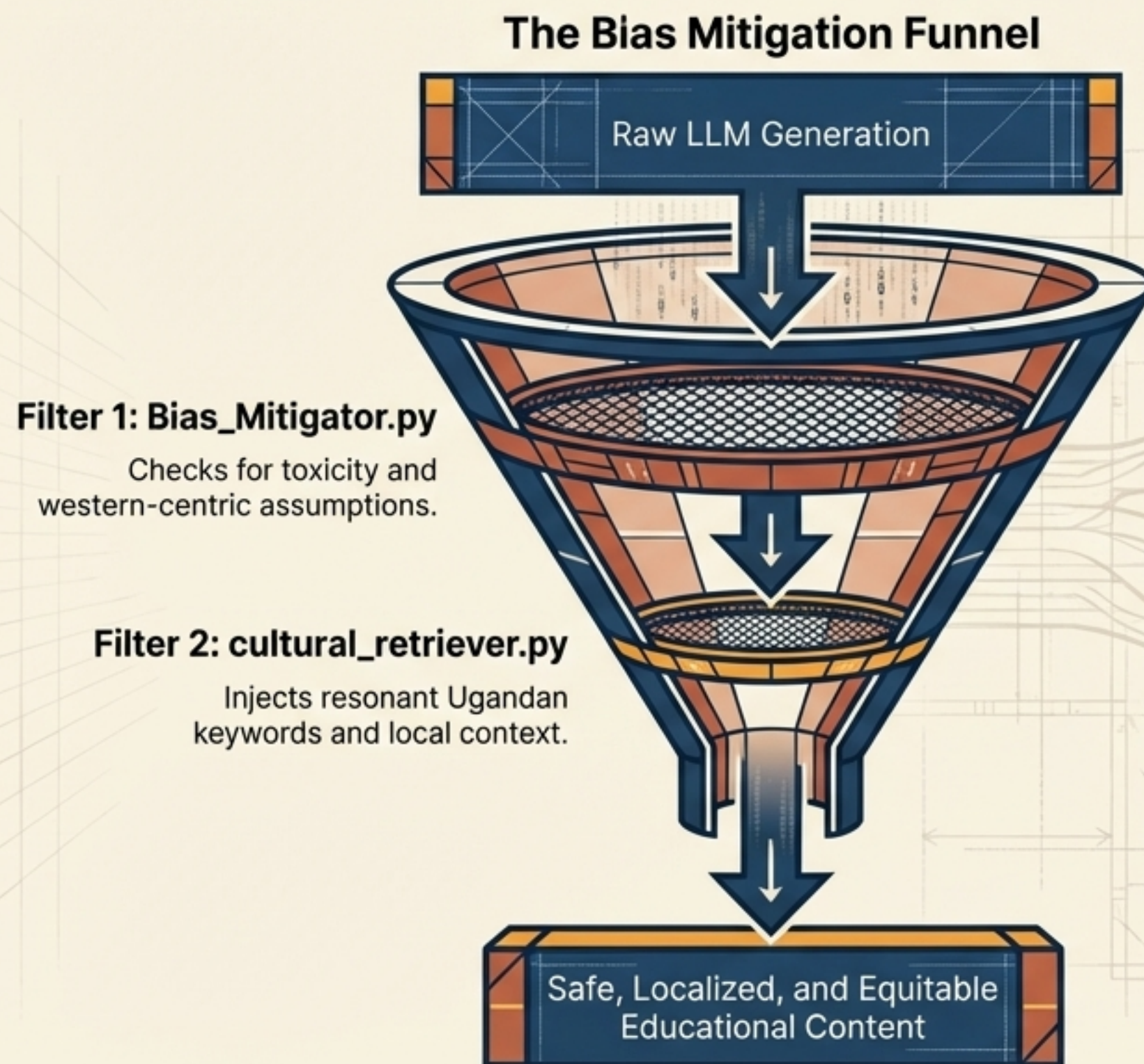


The Foundation: ALETU's Technology Stack



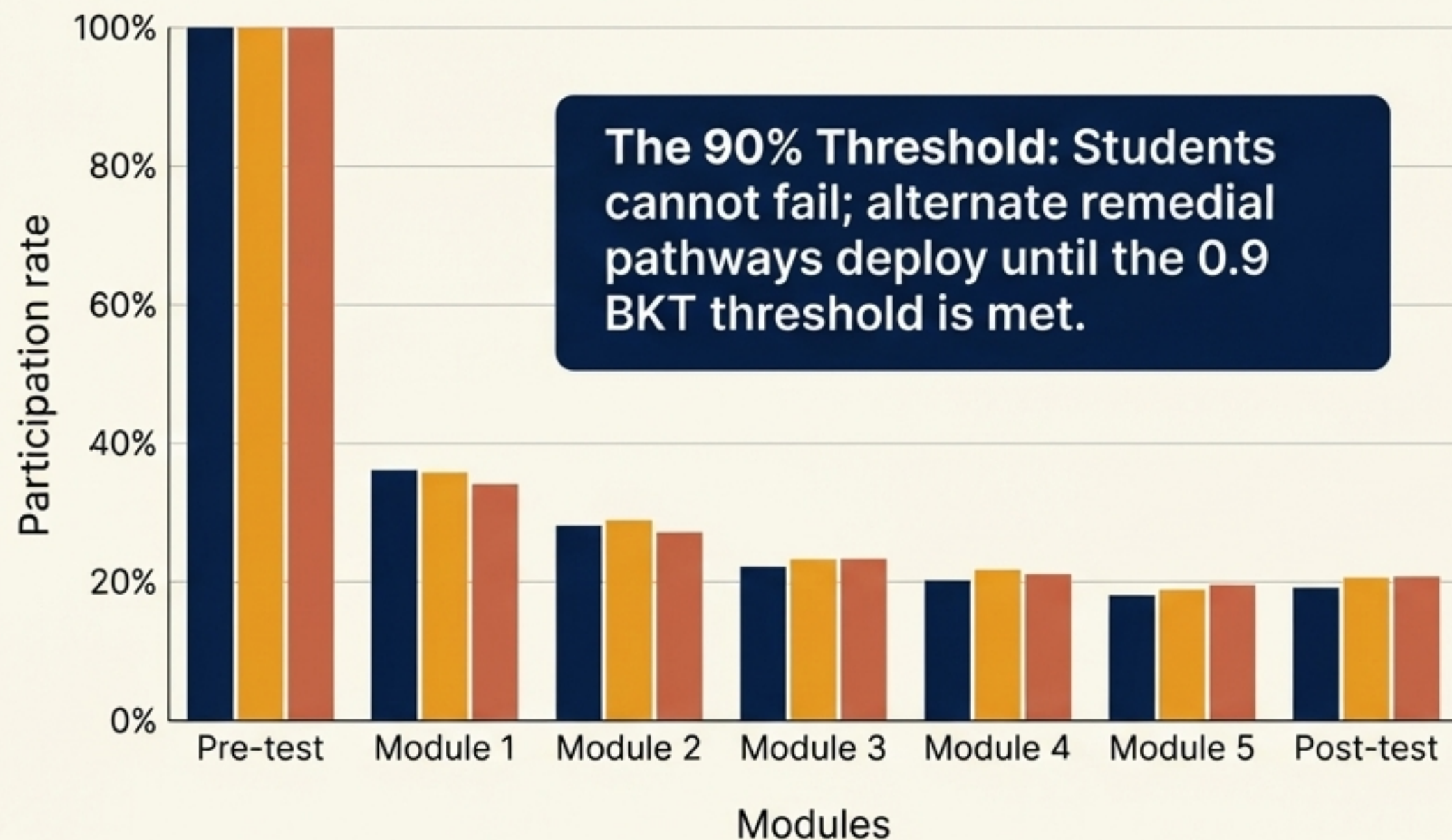
The Soul: Key Values in the Code

Protecting the Learner via Systemic Equity



Privacy First:
PII is hashed;
data is localized
and secure.

The Human Impact: Guaranteeing Mastery



Tracking emotional persistence via
`tracking_emotion_and_motivation.py`

100%

Empowering Educators

The Single Teacher Success Rate (SSR) metric tracks how AI alerts teachers to highly specific human interventions, supporting rather than replacing them.

Leapfrogging to the Future of Pan-African EdTech

Just as Africa bypassed legacy banking with mobile money, ALETU bypasses legacy educational infrastructure with offline-capable, mobile-first AI. Aligned with AUDA-NEPAD's EdTech 2030 Vision and Agenda 2063.

**From a static grid to a neural curriculum.
The revolution is adaptive.**

